Highly-Expressive Spaces of Well-Behaved Transformations: Keeping It Simple

Oren Freifeld[1] Søren Hauberg[2] Kayhan Batmanghelich[1] John W. Fisher III[1]

1: MIT CSAIL, 2:Technical University of Denmark

Massachusetts Institute of Technology

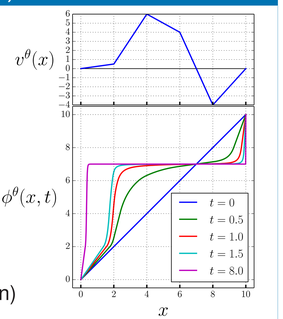
Technical University of Denmark

Introduction

- **Diffeomorphisms on** $\Omega \subset \mathbb{R}^n$
  - $T : \Omega \to \Omega$ is a (C^1) diffeomorphism if: T^{-1} exists; T and T^{-1} are differentiable
  - Diffeomorphisms have **applications** in many fields, including **computer vision**, **medical imaging** and **robotics**; there are also new **machine-learning** applications
- **Problem:** Current representations of highly-expressive diffeomorphisms are **complicated** and **computationally expensive**
- **Goal:** We want a **representation** of diffeomorphisms that combines **expressiveness**, **simplicity** and **practicality**; particularly, want it to lead to simpler probabilistic modeling, rapid likelihood evaluation, and more tractable inference

Proposed Solution: CPA-Based (CPAB) Transformations

- **The rational behind the name CPAB**
  - PA = Piecewise-Affine
  - CPA = Continuous Piecewise-Affine
  - CPAB transformations are obtained by **integrating CPA velocity fields**

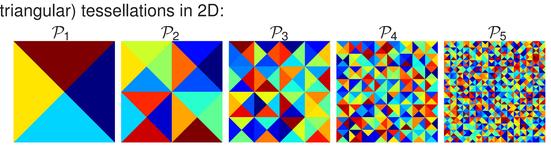


- **Benefits of using CPAB Transformations**
  - Evaluated **fast** with **high accuracy** (exact in 1D)
  - Easy to implement, including in **GPU**
  - Modest mathematical preliminaries
  - **finite dimensional**: $v = v^\theta$, $\theta \in \mathbb{R}^d$, $d < \infty$
  - **continuously-defined** velocity fields (no discretization)
  - **optional constraints** (e.g. volume preservation)
  - useful **modeling** choices (e.g.: smoothness priors, coarse-to-fine)
  - Tractable **inference** (even with, e.g., **MCMC**)

Notation

- $\mathcal{P} = \{U_1, \ldots, U_N\}$, an N-cell tessellation of Ω — N: # of cells

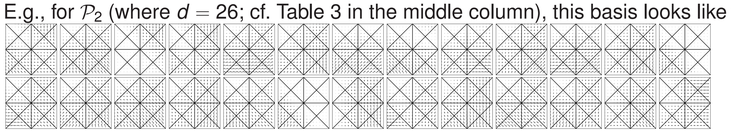
  Examples for (triangular) tessellations in 2D:



- $\gamma(x) = c \Leftrightarrow x$ belongs to U_c — cell index: $c \in \{1, \ldots, N\}$
- $\mathcal{V}'$: The linear space of PA velocity fields — $D \triangleq \dim(V')$
- $\mathcal{V}$: The linear space of CPA velocity fields — $d \triangleq \dim(V)$
- M: The nonlinear space of CPAB transformations — $\dim(M) = d$
- v^θ: an element of $\mathcal{V}$ — $\theta \in \mathbb{R}^d$
- θ: weights in a lin. comb. of CPA fields that form **an** orthonormal basis of $\mathcal{V}$

  This basis corresponds to a "skinny-tall" $D \times d$ orthonormal matrix B

  E.g., for $\mathcal{P}_2$ (where $d = 26$; cf. Table 3 in the middle column), this basis looks like this:



- $T^\theta(x) = \phi^\theta(x, 1)$ where $\phi^\theta : \Omega \times \mathbb{R} \to \Omega$ solves the integral equation

$$\phi^\theta(x,t) = x + \int_0^t v^\theta(\phi^\theta(x,\tau))\, d\tau\,, \quad v^\theta \in \mathcal{V}\,, \tag{1}$$

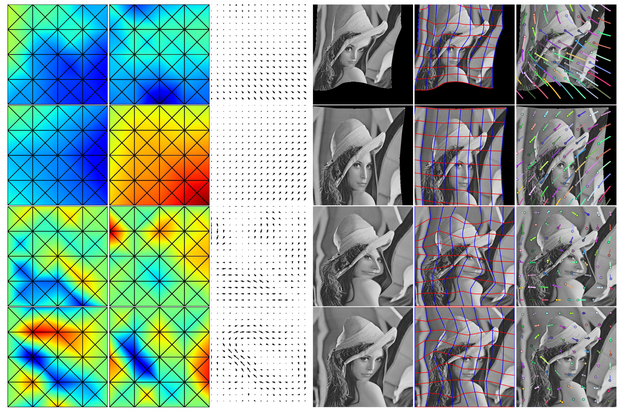
a relation denoted by $T^\theta = \exp(v^\theta)$; i.e.: $M \triangleq \{\exp(v^\theta) : v^\theta \in \mathcal{V}\}$

Smoothness Prior

Two options (with pros and cons to each choice)

1. Let $A = A_1, \ldots, A_N$. Construct a Gaussian smoothness prior on $\mathbf{vec}(A)$, denoted $\mathcal{N}(0_{D\times 1}, \Sigma_{\mathrm{PA}})$, and then project it down, from $\mathcal{V}'$ to $\mathcal{V}$; i.e., $p(\theta) = \mathcal{N}(0_{d\times 1}, B^T \Sigma_{\mathrm{PA}} B)$
2. Use any prior that penalizes differences between velocities at adjacent vertices

Example Application: Image Warping



4 random warps, from top to bottom: unconstrained; w/ volume preservation (VP); w/ boundary condition (BC); w/ BC & VP.

The heat maps (blue: negative; green: 0; red: positive) show the horizontal (col 1) and vertical (col 2) of v^θ (col 3, shown in select location) are CPA. Cols 4-6 illustrate the warp, deformation, and (select) ODE trajectories

CPAB Formula

- Velocity fields are widely used to overcome the **nonlinearity of spaces of diffeomorphisms**; their entire integration, however, is done via numerical approximations (time/accuracy tradeoff). Unlike with other spaces of highly-expressive velocity fields, integration of CPA velocity fields is given in **either closed form (in 1D) or "essentially" closed form (for** $n > 1$**)**,

$$\phi^\theta(x,t) = (\psi^{t_m}_{\theta,c_m} \circ \ldots \circ \psi^{t_2}_{\theta,c_2} \circ \psi^{t_1}_{\theta,c_1})(x) \tag{2}$$

  where $t \in \mathbb{R}$ and $\psi^t_{\theta,c}(\cdot) : \Omega \to \mathbb{R}^n$ is affine, invertible, and orientation preserving
- This compact notation, however, hides an ugly truth:

$$\phi^\theta(x,t) = (\psi^{t_{m_{x,\theta,t}}(x,\theta,t)}_{\theta,c_{m_{x,\theta,t}}(x,\theta,t)} \circ \ldots \circ \psi^{t_2(x,\theta,t)}_{\theta,c_2(x,\theta,t)} \circ \psi^{t_1(x,\theta,t)}_{\theta,\gamma(x)})(x) \tag{3}$$

- In 1D, the RHS quantities have closed form
- For $n > 1$, we propose a **fast and highly-accurate specialized solver** which alternates between the exact solution and a generic numerical solver

Dimensionality, Timings, and Accuracy

1D

N	5	25	45	65	85	105
our proposed solver	0.06	0.08	0.08	0.09	0.11	0.12
a generic solver	0.33	0.43	0.41	0.43	0.47	0.48

Table 1: Average timings, in [sec], for integration of random CPA fields ($N_{\mathrm{pts}} = 10^5$; $N_{\mathrm{steps}} = 100$; $n_{\mathrm{steps}} = 10$)

N	5	25	45	65	85	105
improvement [%]	2.99	12.56	11.26	9.98	3.29	3.73

Table 2: Accuracy: Average improvement, in %, of the proposed solver ($N_{\mathrm{steps}} = 10$; $n_{\mathrm{steps}} = 10$) over the generic solver (100 steps). The fact that the error is computable, is by the virtue of our closed-form solution

2D

$\mathcal{P}$	N	N_v	$D = 6N$	$d = 2N_v$
$\mathcal{P}_1$	4	5	24	10
$\mathcal{P}_2$	16	13	96	26
$\mathcal{P}_3$	64	41	384	82
$\mathcal{P}_4$	256	145	1536	290
$\mathcal{P}_5$	1024	1025	6144	1090

Table 3: Values of N, N_v, $D = \dim(\mathcal{V}'_{\Omega,\mathcal{P}})$, and $d = \dim(\mathcal{V}_{\Omega,\mathcal{P}})$ for the above tessellations

$\mathcal{P}$	$\mathcal{P}_1$	$\mathcal{P}_2$	$\mathcal{P}_3$	$\mathcal{P}_4$	$\mathcal{P}_5$
our proposed solver	0.06	0.09	0.15	0.24	0.27
a generic solver	0.22	0.23	0.25	0.26	0.28

Table 4: Average timings, in [sec], for integration of random CPA fields. Image size is 512 × 512, $N_{\mathrm{steps}} = 100$ and $n_{\mathrm{steps}} = 10$

3D

100^3 voxels, with $N = 320 = 4^3 \times 5$ tetrahedra ($N_v = 125$) + boundary conditions (so $d = 225$): ~ 0.2 [sec] on average

Let's Eliminate Some Possible Sources of Confusion

- CPAB transformations are **not affine** (due to the x-dependencies in Eqn. (**??**)), **not CPA** (since they are not PA), and **not piecewise quadratic** (Eqn. (**??**) should not be confused with $y \mapsto \int_{0_{n\times 1}}^{y} v^\theta(x)\, dx$)
- $x \mapsto \phi^\theta(x, t = 1)$ should not be confused with $x \mapsto x + v^\theta(x)$ (a non-diffeomorphism, **optical-flow-like** representation), the latter being only a **Taylor approximation** of the former. Thus, our use of flexible parametrized vector fields is different from, e.g., [Nir et al., IJCV '08] or [Wulff et al., CVPR '15]

Additional Select Applications

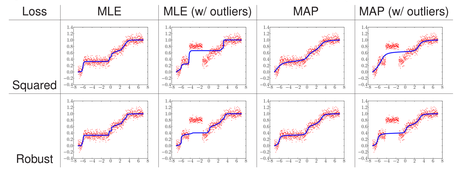
Monotonic Regression

Let J be an interval

Find an increasing $F^\theta : J \to \mathbb{R}$ s.t. $\{\mathrm{dist}(F^\theta(x_i), y_i)\}$ are small

$F_0 : J \to \mathbb{R}$ is $\uparrow \Rightarrow$ so is $F^\theta \triangleq T^\theta \circ F_0 : J \to \Omega$

F_0, not shown, is a straight line from (-7,0) to (7,1)



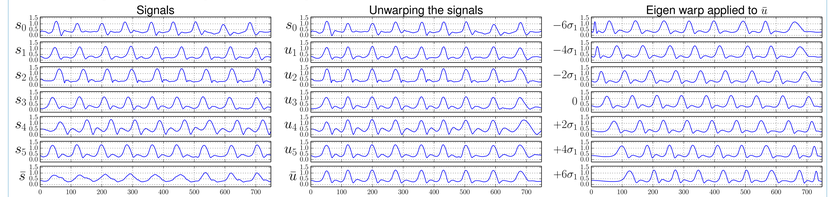
3D Shape Warping



Left: source shape (from `http://tosca.cs.technion.ac.il`)

Right: Applying a random fixed-boundary volume-preserving CPAB transformation to it. $N = 320$ (4^3 cubes, each one split into 5 tetrahedra) and $N_v = 125$. Here $d = 20$ (and not $375 = 3N_v$) due to the constraints

Time Warping: Analysis and Synthesis

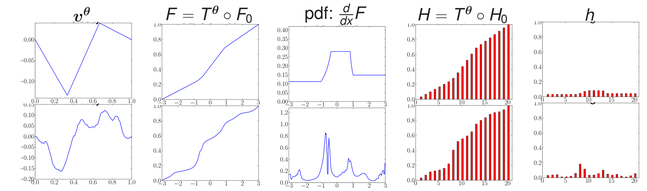


Modeling CDF/histogram-valued Data

F_0, not shown, is a CDF of the uniform distribution on $J = [-3, 3]$

H_0, not shown, is a cumulative sum of a 20-bin uniform histogram

Note that $H = T^\theta \circ H_0$, a cumulative sum of a new histogram h



Learned Data Augmentation

[Hauberg, Freifeld, Larsen, Fisher and Hansen, arXiv:1510.02795]

Dataset	MLP Test error	ConvNet Test error
MNIST	$1.42\% \pm 0.055$	$0.65\% \pm 0.08\%$
InfiMNIST	$0.89\% \pm 0.079$	$0.49\% \pm 0.04\%$
AlignMNIST	$0.58\% \pm 0.022$	$0.44\% \pm 0.02\%$
InfiMNIST500	$2.59\% \pm 0.050$	$1.04\% \pm 0.07\%$
AlignMNIST500	$1.06\% \pm 0.047$	$0.84\% \pm 0.05\%$

Table 5: Test error on the different datasets

Prettify/Exploit Dense correspondences



First computed siftFlow [Liu et al.]; next, discarded the original images and, treating siftFlow as noisy correspondences, inferred a non-stationary v^θ, piecewise-constant w.r.t. t, in a greedy way

Additional Applications Include:

Image registration, nonlinear ICA, warped Gaussian processes, diffeomorphic mixture models

Remarks:

- More strictly, $\mathcal{V}'$, $\mathcal{V}$ and M depend on Ω and $\mathcal{P}$; i.e., $\mathcal{V} = \mathcal{V}_{\Omega,\mathcal{P}}$ and $M = M_{\Omega,\mathcal{P}}$
- The intervals/triangles/tetrahedra from the paper generalize to convex n-polytopes; our code also supports n-orthotopes (including for $n > 3$) as well as another type of a basis; each choice of cell type and basis has pros and cons
- In principle we can also use an adaptive (data-dependent) $\mathcal{P}$ but then B will have to be recomputed on the fly; we thus opt to use fixed-but-fine tessellations an let the smoothness prior handle no-data regions
- The novelty is not in using finite-dimensional spaces of velocity fields to generate diffeomorphisms (cf., e.g., splines or [Arsigny et al.]) but in picking a particular space with **all** the benefits mentioned earlier
- we focused on the stationary case; in the non-stationary case some of the benefits will be preserved, but the proposed solver will be inapplicable; however, in one non-stationary case we can (trivially) use this solver as is: when θ is piecewise constant w.r.t. t

For more details, paper and code: `http://people.csail.mit.edu/freifeld/publications`